Greecon
SH.P.K

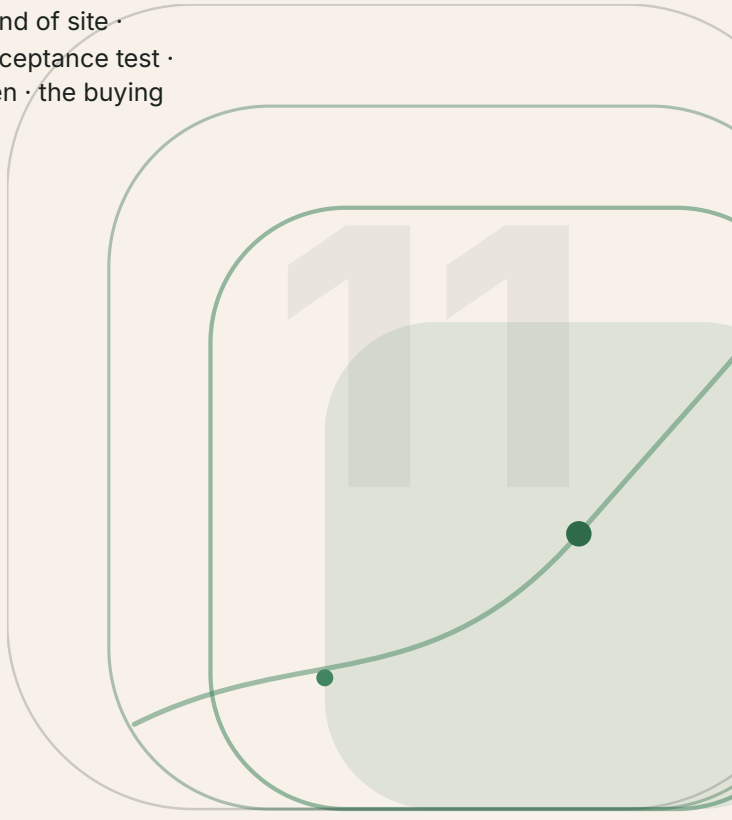
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HARDWARE SHEET HW11 · QUALIFIED BY SPECIFICATION

Pressure transmitter

The sensor that says what the pressure in the pipe is. It sits on the pipe behind an isolating valve and sends a 4 to 20 mA signal.

QUALIFIES	Qualified by specification. Any 4 to 20 mA pressure transmitter with stainless wetted parts and a range that suits the pipe.
USED IN	Kits H, F and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who chooses its range at the survey, the plumber who fits it and the electrician who wires it
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW11

Pressure transmitter

The pressure transmitter says what the pressure in the pipe is. It is screwed into the pipe behind an isolating valve, so that it can be taken out without draining the pipe, and it sends a current between 4 and 20 mA on two wires: 4 mA at no pressure and 20 mA at the top of its range. The 4 to 20 mA module (sheet HW08) reads that current on its input 2, and Greecon asks for one number: the top of the range on the transmitter's label.

At a glance

WHAT IT DOES	Measures the pressure in the pipe, through an isolating valve so that it can be removed without draining.
HOW IT QUALIFIES	**Qualified by specification.** Any 4 to 20 mA pressure transmitter with stainless wetted parts and a range that suits the pipe.
SPECIFICATION	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts
WIRED TO	Input 2 of the 4 to 20 mA module (sheet HW08), in a loop with the 24 V supply.
FITTED BY	A plumber makes the wet connection, through an isolating valve. The electrician wires its loop.
PLANNING ESTIMATE (EUR)	40 to 150. Not a quote.
USED IN	Kits H, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. The pipe decides what is bought: its pressure range is asked at the survey.

It qualifies if:

- ☐ It has a two-wire 4 to 20 mA output.
- ☐ **Its range suits the pipe.** The top of its range is above the highest pressure the pipe sees. 0 to 10 bar is the usual one.
- ☐ It measures gauge pressure: zero in the open air, like the pressure gauge on the line that it is checked against.
- ☐ Its wetted parts are stainless steel. On a drinking-water pipe, ask the plumber whether the transmitter and its fitting may be used there.
- ☐ Its thread is known before the visit, so that the plumber brings the right fitting and the isolating valve.

- [ ] Its label gives its range, in bar. Photograph the label.

STOP

Do not fit it without an isolating valve. Without one the transmitter cannot be taken out, checked or changed without draining the pipe. And do not buy the kind with a voltage output (0 to 10 V): the module reads current.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One, on the pipe the pump feeds. It gives Line pressure.
S Solar plant with grid connection	NOT USED	None	A solar plant has no water side.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, on the irrigation line.
W Water tank and pump station	IN THE KIT	1	One, on the pipe the pump feeds. The pipe's pressure range, asked at the survey, decides the transmitter's range.

What Greecon reads from it

One reading: **Line pressure**, in bar. 4 mA is 0 bar, and 20 mA is the top of the range.

The label says	4 mA	12 mA	20 mA	With a wire off, Greecon shows
0 to 10 bar	0 bar	5 bar	10 bar	-2.5 bar, and says that it cannot be a line pressure
0 to 6 bar	0 bar	3 bar	6 bar	-1.5 bar, and says the same
0 to 4 bar	0 bar	2 bar	4 bar	-1.0 bar, which Greecon accepts as a pressure. See *What goes wrong*

A pressure outside -1 to 40 bar is called a reading that cannot be real. There is **no standard alert** on the pressure: Greecon shows it and its trend.

Settings, and where they are made

Nothing is set on the transmitter: its range is fixed when it is made.

What	Value	Where it comes from
The top of the range	In bar: 10, for a 0 to 10 bar transmitter	The label
Its input on the module	Input 2	Wired by the electrician

Fitting

A plumber makes the wet connection, and **ELECTRICIAN** wires the loop. **PROJECT MANAGER** watches, photographs and writes down. The pipe is the customer's: it is opened by the customer's plumber, or by one the customer agrees to.

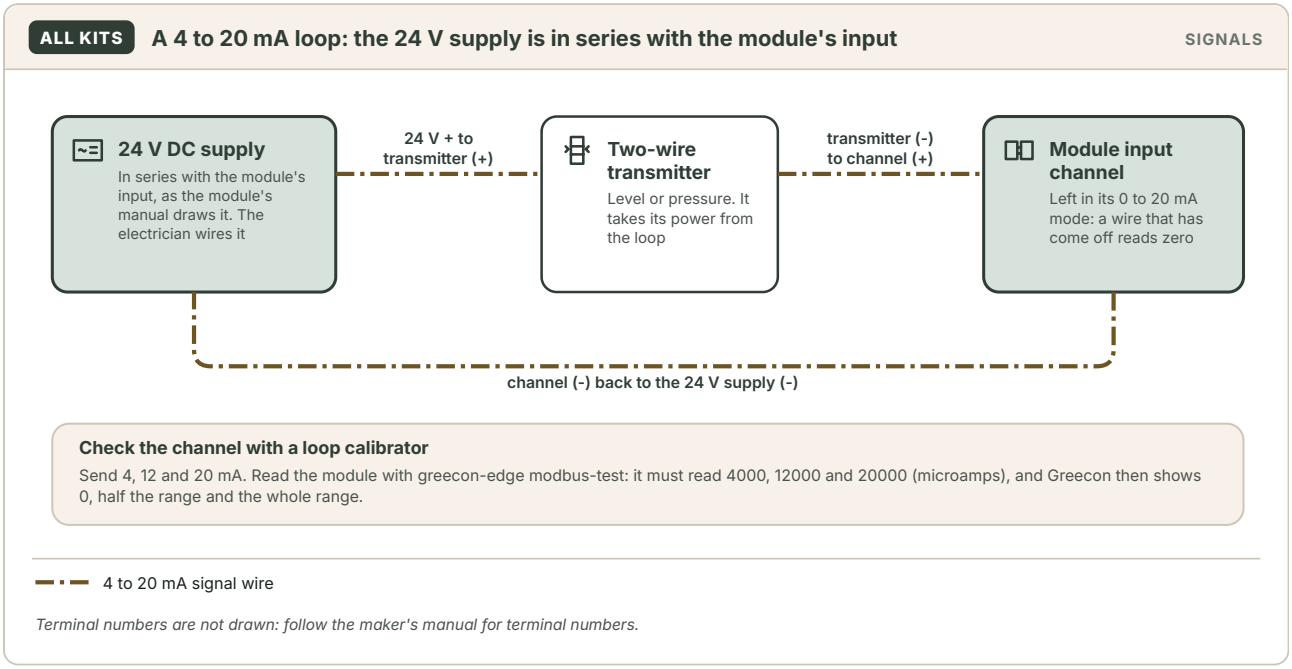


Figure 4-3. A 4 to 20 mA loop: the 24 V supply in series with the module's input.

- ☐ The transmitter goes on the pipe the pump feeds, **through an isolating valve.**
- ☐ Where there is a pressure gauge on the line already, the transmitter goes near it, so that the two can be compared.
- ☐ The loop is wired to **input 2** of the module, with the 24 V supply in series, as the module's manual draws it.
- ☐ The two wires are numbered at both ends. The label, the valve and the terminals are photographed.

In the platform

The transmitter is not a device in the platform. It is one reading of the **Water I/O module**, which the box finds in step 4, **Where each device is on the network**.

- **Its range.** In step 5, **Which register holds which reading**, under **Pressure transmitter**, give the **Top of the range** (10) and press **Save the ranges**. Greecon works out the register and the scale: nobody types either.
- **A number is read the way people write it:** "2,5" is 2.5.

Acceptance test

At the office, before the kit leaves:

- ☐ The label is photographed, and the top of its range is written on the install card.
- ☐ With a loop calibrator on input 2 in place of the transmitter, the module reads 4000, 12000 and 20000 (sheet HW08).
- ☐ In its loop on the desk, in the air, the transmitter reads about 4000: 4 mA, no pressure.

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 1
```

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Pressure. Against the pressure gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
A pressure below zero that never moves (-2.5 with a 0 to 10 bar transmitter)	No current in the loop: a wire has come off, or the loop has lost its 24 V supply. With a transmitter of 0 to 4 bar or less the number is between -1 and 0, and Greecon does not call it impossible	Read input 2 from the box: 0 means no current. The electrician checks the loop and its supply
Greecon and the gauge agree at rest and differ under pressure	The top of the range was given wrongly (a 0 to 16 bar transmitter given as 10)	Read the label again and give the range again
Zero in Greecon while the gauge shows pressure	The isolating valve is closed	The plumber, or the customer's own person, opens it
The right number on the wrong reading	The transmitter is on another input	Pressure is on input 2

More is in the runbooks (Project Manager Field Manual, PDF 02): R3 (a reading looks wrong) and R18 (a reading is impossible after the ranges were given).

Not yet proven

- This software has not yet read a pressure transmitter through a real 4 to 20 mA module. The scale is tested against a pretend module. The first real one is checked against the gauge on the line, and both numbers are written down.

- The prices are planning figures from shop listings seen in October 2026, not quotes. A known make of pressure transmitter was seen from GBP 129.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	